

Summary

Reference: RA130876/1

Sign-off Status: Authorised

Date Created:	12/05/2026	Confidential?	No
Assessment Title: AI-assisted interpretation of morphological features in British butterflies using museum specimens			
Assessment Outline: Natural history collections contain extensive material that underpins biodiversity research, yet morphological identification can remain challenging for closely related taxa, even for experienced specialists. Recent advances in computer vision and explainable artificial intelligence (AI) provide an opportunity to systematically investigate which morphological features are most informative for taxonomic discrimination. This project will focus on British butterfly material held within museum collections. The student will work with a small number of closely related taxa or populations that are known to be visually challenging to distinguish using traditional approaches. Using high-quality specimen images captured during the project or drawn from existing digitised collections, the student will build and curate an image dataset suitable for machine learning analysis. Following established workflows developed in recent literature, the project will apply convolutional neural networks and explainable AI methods to identify, visualise, and extract diagnostically informative morphological features. These features will then be analysed and interpreted in a biological context, with reference to existing identification literature and expert knowledge. The project will emphasise the relationship between AI model decisions and underlying morphology, with the aim of improving understanding of which visual traits are most informative for discrimination between closely related butterfly taxa. This project is suitable for students with a background in ecology, zoology, biodiversity science, computer science, or data science. Prior experience with programming or machine learning is advantageous but not essential, as training and guidance will be provided. While this project is framed around British butterflies, there is scope for alternative versions focusing on different insect groups or other taxa, depending on student interest, material availability, and supervisory capacity. The core methodological approach is transferable across multiple taxonomic contexts.			
Area Responsible (for management of risks)		Location of Risks	
Division, School, Faculty, Institute:	External Organisations affiliated to UCL	Building:	Off-Site
Department:	UCL Partners	Area:	
Group/Unit:	All Groups/Units	Sub Area:	
Further Location Information:	8th floor common room of Natural History Museum & Marshgate Building, 7 Sidings St, London E20 2AE		
Is additional GM or HG approval required? Only relevant to specialist biological risk assessments (GMM2, GMM3, HG2, HG3, GM animals, GM plants) except GMM class 1.:	Click SELECT to change ONLY if this is a GMM Class 2, GMM Class 3, HG2, HG3, GM animals or GM plants risk assessment		
Assessment Start Date:	26/06/2026	Review or End Date:	15/10/2026
Country Location:	GMM Class 2, GMM Class 3 or Hazard Group 3 Risk Assessment :		
Relevant Attachments:			
Description of attachments:			
Location of non-electronic documents:			
Assessor(s):	Zhao, Zeyu		
Approver(s):	Hayley Boakes James Labisko		



Risk Assessment

Signed Off:	James Labisko (23/06/2026 09:01) Hayley Boakes (26/06/2026 15:09)
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PEOPLE AT RISK (from the Activities covered by this Risk Assessment) *

CATEGORY
Post-Graduates

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1. Taking photos of specimens in NHM

Description of Activity:	This activity involves travelling from UCL to the Natural History Museum (NHM) in South Kensington for the purpose of photographing entomological specimens as part of a postgraduate research project on butterfly classification. My supervisor Jack Hollister will provide guidance regarding specific procedures at common room on 8th floor.
List those managing this Activity and their competence:	

Hazard 1. Ergonomic strain from sustained fine-focus photography		
The photography work requires prolonged periods of looking through a camera viewfinder or at a live-view screen while making fine adjustments to focus, aperture, and lighting. This involves maintaining a static, bent-forward posture for extended durations (e.g., 30–60 minutes without break), which can cause neck, shoulder, and lower back discomfort. Additionally, repeated fine motor movements of the fingers and wrists when adjusting camera settings or triggering the shutter may lead to temporary muscle fatigue or strain. No heavy lifting or specimen transfer is involved.	Existing Control Measures I will take short breaks every 20–30 minutes to stand up, stretch my neck and shoulders, and look away from the viewfinder. I will also vary tasks: after every 10–15 specimens, I will step back, check images on the laptop screen, and reorganise the workspace, which naturally breaks prolonged static posture. If I feel any persistent pain or numbness, I will stop immediately, inform the museum staff, and seek advice from my supervisor or UCL Occupational Health.	Further Control Measures

Hazard 2. Slip, trip, and general laboratory risks (without specimen transfer)

The photography area may contain trailing cables (from camera power supplies, laptop chargers, external lights), tripod legs, and unattended bags. These can create trip hazards. Additionally, the reading room may have low lighting in some areas, increasing the risk of missing a step or bumping into furniture. There is also a low risk of allergic reaction to dust or mould from old specimen drawers (e.g., respiratory irritation or skin contact).

Existing Control Measures

Lighting: I will use the museum's overhead lights and, if permitted, a small portable LED lamp for focusing. I will ensure that any additional lighting does not create glare or shadows that could cause me to misjudge distances.

Dust/mould awareness: I will avoid touching the specimens directly and will not blow on them. If I notice visible mould or excessive dust, I will not open that drawer and will report it to the curator. I will wash my hands after each session and avoid touching my face during work. I have no known allergies to museum pests or mould

Emergency procedures: I have noted the location of the nearest fire exit, first aid kit, and telephone. In case of any incident, I will follow the museum's emergency protocols and contact UCL's 24/7 support if necessary.

Further Control Measures



Risk Level

With Existing Controls:

Risk Level **B - Low / Tolerable** Likelihood Severity

2. Desk Work

Description of Activity:

This activity covers all desk-based work required for my Master's dissertation, including literature review, data processing, machine learning model training, and thesis writing. The work is mainly carried out in my office at UCL or at my home desk. I will work for approximately 6–8 hours per day, using a computer, keyboard, and mouse. The main risks are musculoskeletal problems (e.g. back, neck, wrist pain) from poor posture, visual fatigue from prolonged screen use, and mental health issues related to sedentary behaviour and social isolation.

List those managing this Activity and

their competence:

Hazard 1. DSE-related musculoskeletal and visual fatigue, plus sedentary behaviour

Prolonged use of display screen equipment with poor ergonomics can lead to upper back, neck, shoulder, or wrist discomfort, as well as eye strain, headaches, and blurred vision. Sitting for long periods without breaks is also linked to lower back pain and may contribute to increased stress, anxiety, or low mood during intense research periods.

Existing Control Measures

I take short breaks every 50–60 minutes (e.g., standing up, walking around). I follow the 20-20-20 rule for eye health: every 20 minutes, I look at something about 6 metres away for at least 20 seconds. I will ensure my chair provides lumbar support, my feet rest flat on the floor, and the screen is at arm's length with the top of the screen at or below eye level. The keyboard and mouse are positioned so that my wrists are straight. I will monitor any persistent pain or fatigue and report it to my supervisor. To reduce sedentary time, I use stairs instead of lifts, walk during lunch breaks, and have regular in-person or online meetings with peers to maintain social contact. I also maintain a clear boundary between work hours and personal rest.

Further Control Measures

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Risk Level

With Existing Controls:

<u>Risk</u>	A - Very Low / Trivial	<u>Likelihood</u>
<u>Level</u>		<u>Severity</u>

Actions

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Actions associated with this Risk Assessment	
*** No Actions have been recorded***	